

PROG2053 – Web Technologies

Assignment 2: Basic JavaScript and the browser

Instructions

If any part of this assignment is unclear, students are expected to make reasonable assumptions and clearly document them within their submission (e.g., as comments in the code or in an accompanying note). All stated assumptions will be considered during the marking process.

This assignment should be delivered as a .txt file.

Q1. JavaScript Syntax, Operators, and Loops

```
let total = 5;
```

```
for (let j = 3; j <= 8; j++) {  
  if (j % 2 === 0) {  
    total += j;  
  } else {  
    total *= j;  
  }  
}
```

1. What will be the final value of **total** after the code executes?
2. What is the purpose of the condition **j % 2 === 0** in this code?
3. Rewrite the same logic using a **while loop** instead of a **for loop**.
4. Explain in your own words how **operators** (+, *, %) are used in this example.

Q2. Arrays, Strings, and Classes

Part A – Strings and Arrays

1. Convert the string *I am writing in Javascript!* into an array.
 - 1.1. **Explain** which JavaScript method is used and why.
2. Suppose this string was already stored as an array. How would you **convert** it back into a single string?
 - 2.1. **Explain** the method used for joining array elements.

Part B – Constructor Function and Classes

Rewrite the following constructor function as a JavaScript class and ensure that the method **describe** is properly implemented as part of the class definition.

Use different variable names if needed.

```
function Food(name, cooktime) {
  this.name = name;
  this.cooktime = cooktime;
}
Food.prototype.describe = function() {
  console.log(`${this.name} is delicious and will take ${this.cooktime} to
finish cooking.`);
};
const food1 = new Food('Pizza', '15 min');
food1.describe(); // Expected output: "Pizza is delicious and will take 15
min to finish cooking."
```

Q3. Working with Objects and String Methods

Create an object named student with the following properties:

- name, age, course, and a greet() method that returns “Hello, I am [name]”.

Display the greeting inside the webpage using **innerHTML**.

Create a **string** variable and use string methods to **display** different versions of the text:

- .toUpperCase()
- .length
- .pop()
- .replace()

Q4. Conditional Operators

Ask the user to enter the current outdoor temperature in degrees and check whether it is hotter than, colder than, or exactly equal to a perfect target temperature value set in your code.

Output the result to the user.

Q5. Classes, Promises

Define a **JavaScript class** named Book with:

- Properties: name, author, ISBN
- A **getter** that returns a description like “The book is called The Hunger Games by Suzanne Collins and has ISBN: 978-0-439-02352-8.”
- A **setter** to change the name.

Create a **Promise** that resolves after 3 seconds with a message "Data Loaded Successfully!".

- Use **then()** and **catch()** to display the message in the console.

Display the book description on the webpage once the Promise resolves.

Q6. DOM Traversal Practice (Nested Sections)

HTML Structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>DOM Traversal Exercise</title>
</head>
<body>

<section id="main-section">
  <h2 id="main-title">Welcome to JavaScript</h2>
  <section id="nested-section">
    <p id="intro-text">This is an introduction paragraph inside a nested
section.</p>
    <!-- Comment node here -->
    <p id="nested-detail">Details paragraph inside nested section.</p>
  </section>
  <p id="footer-text">This is a footer paragraph outside the nested
section.</p>
</section>

<script>
const introText = document.getElementById('intro-text');

// 1. parentNode of introText
console.log('parentNode:', introText.parentNode.nodeName);

// 2. childNodes of parentNode (includes text and comment nodes)
console.log('childNodes:', introText.parentNode.childNodes.length);

// 3. firstElementChild of parentNode
console.log('firstElementChild:',
introText.parentNode.firstElementChild.nodeName);

// 4. nextSibling of introText
console.log('nextSibling:', introText.nextSibling.nodeName);

// 5. nextElementSibling of introText
console.log('nextElementSibling:', introText.nextElementSibling.nodeName);
</script>

</body>
</html>
```

Tasks:

1. Represent the **full DOM tree** for this HTML, showing the hierarchy of elements and text/comment nodes.

2. Explain the following DOM traversal operations:

- Identify which element or node is being accessed.
- Name the DOM traversal property used.
- Explain how the property works and why it returns that result in this context.

Q7. Functions and Events

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">

  <meta name="viewport"
        content="width=device-width, initial-scale=1.0">

  <title>Course Registration</title>
</head>

<body>
  <h1>Course Registration</h1>

  <label for="course">Enter a course name:</label>
  <input type="text" id="course">

  <button id="register-button">Register</button>

  <p id="message"></p>

  <script>
    // Write your JavaScript here
  </script>
</body>
</html>
```

1. Create a function named **registerCourse()** that reads and checks the value entered in the input field.
2. Add a **click** event listener to the **Register** button that calls the **registerCourse()** function.
3. If the input field is empty, display: **Please enter a course name.**
4. Otherwise, display: **You are registered for [course name].**
5. Explain:
 - What the **click** event does.
 - What the **registerCourse()** function does.
 - Why **addEventListener()** is used.